



PYTHON BOOTCAMP

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FUNCTIONS



Functions:

- Reusable block of code that do specific task

```
3  # Functions with parameters
4  def greet_person(name):
5      |   print(f"Hello, {name}!")
6
7  greet_person("Alice")
8
9  # Functions with return values
10 def add_numbers(a, b):
11     |   return a + b
12
13 result = add_numbers(5, 3)
14 print(result) # 8
15
16 # Default parameters
17 def greet_with_title(name, title="Mr."):
18     |   return f"Hello, {title} {name}!"
19
20 print(greet_with_title("Smith"))           # "Hello, Mr. Smith!"
21 print(greet_with_title("Johnson", "Dr. ")) # "Hello, Dr. Johnson!"
```

FUNCTIONS



args :

- access by index: args[0]
- unpacking: func(*list)

```
23 # *args - variable number of arguments
24 def sum_all(*args):
25     return sum(args)
26
27 print(sum_all(1, 2, 3, 4, 5)) # 15
```

kwargs :

- access by key: kwargs['key']
- unpacking: func(**dict)

```
29 # **kwargs - keyword arguments
30 def print_info(**kwargs):
31     for key, value in kwargs.items():
32         print(f"{key}: {value}")
33
34 print_info(name="Alice", age=25, city="New York")
```

FUNCTIONS



args & kwargs :

```
36 # Combining *args and **kwargs
37 def flexible_function(*args, **kwargs):
38     print("Positional arguments:", args)
39     print("Keyword arguments:", kwargs)
40
41 flexible_function(1, 2, 3, name="Alice", age=25)
```

FUNCTIONS



lambda :

- anonymous function
- small function

```
43 # Lambda functions (anonymous functions)
44 square = lambda x: x**2
45 print(square(5)) # 25
46
47 add = lambda x, y: x + y
48 print(add(3, 4)) # 7
```

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Exercises :

1. Write a function that checks if a number is prime.
2. Build a temperature converter function. (Celsius to Fahrenheit)